

should stretch the limits of our world's computing resources. More sophisticated solutions could rely on logic to cut down the required runtime.

The n queens completion problem and sudoku share another important trait: solutions are very easy to check. That is, if I show you a chessboard with queens on it, it will probably take you only a few moments to check whether you're looking at a solution to the n queens completion problem, and if I show you a grid of 81 numbers, you can easily tell whether you're looking at a correct sudoku solution. The ease with which we can check solutions is, tragically, not matched by the ease of generating solutions—it can take hours to solve a difficult sudoku puzzle that then takes only seconds to verify. This generation/verification effort mismatch is common in many areas of life: I can tell with very little effort whether a meal is delicious, but creating a wonderful meal takes a much greater investment of time and resources. Similarly, I can check whether a painting is beautiful in much less time than it takes to create a beautiful painting, and I can verify whether a plane can fly with much less effort than it takes to build a flying plane.

Problems that are difficult to solve algorithmically but whose solutions are easy to verify are extremely important in theoretical computer science, and they are the deepest and most pressing mystery in the field. Especially courageous adventurers may dare to plunge into these mysteries—but beware the perils awaiting you there.

Solving the Deepest Mysteries

When we say that sudoku solutions are easy to verify but hard to generate, what we mean in more formal terms is that solutions can be verified in *polynomial time*; in other words, the number of steps required for solution verification is some polynomial function of the size of the sudoku board. If you think back to Chapter 4 and our discussion of runtimes, you'll remember that even though polynomials like x^2 and x^3 can grow fast, they are quite slow compared to exponential functions like e^x . If we can verify an algorithmic solution to a problem in polynomial time, we regard that verification as easy, but if the generation of a solution takes exponential time, we regard it as hard.